**DIPLOMA WING**
**RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL**
DIPLOMA IN COMPUTER SCIENCE & ENGINEERING (C04)

*SEMESTER VI*

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| COURSE TITLE | : | MOBILE COMPUTING |
| PAPER CODE | : | 7497 |
| SUBJECT CODE | : | 611 |
| TREORY CREDITS | : | 03 |
| PRACTICAL CREDITS | : | 02 |

**Course Learning Objectives:**

To teaches how to build mobile apps for Android. Students are expected to work on a project as part of the course.

**Course Content:**

**UNIT 1:**

A brief history of Mobile, Types of mobile phone generations, The Mobile Ecosystem, Types of Mobile Applications, Mobile Information Architecture Android Versions, Features of Android, Android Architecture, Installing Android SDK Tools, Configuring Android in Eclipse IDE, Android Development Tools (ADT), Creating Android Virtual Devices (AVD)

**UNIT 2:**

Creating first android application, Anatomy of android application, Deploying Android app on USB connected Android device, Android application components, Activity life cycle, Understanding activities, Exploring Intent objects, Intent Types, Linking activities using intents

**UNIT 3:**

Fragments life cycle, Interaction between fragments, Understanding the components of a screen (Layouts), Adapting to display orientation, Action Bar, Views(UI Widgets)-Button, Toast, ToggleButton, CheckBox, RadioButton, Spinner, WebView, EditText, DatePicker, TimePicker, ListView, ProgressBar, Analog and Digital clock, Handling UI events, List fragment, Dialog fragment

**UNIT 4:**

Menus-Option, Context, Popup, Images-ImageView, ImageSwitcher, AlertDialog, Alarm manager, SMS, E-mail, Media Player, Using camera, recording video, Handling Telephony Manager

**UNIT 5:**

Storing the data persistently-Data Storage Options: preferences, Internal Storage, External Storage, Content Provider , The SQLite database, Connecting with SQLite database and operations-Insert, Delete, Update, Fetch, Publishing android applications, Deploying APK files

**Suggested Lab Work:**

This is a skill course. Topics/tools taught in the class should be practiced in the Lab same week and practiced regularly during the semester till student becomes confident about it. Students should explore features of various tools/technologies introduced during the course and become comfortable with their use. Teacher should give weekly practice tasks as assignment. Learnings from this course should be used in the project/software built.

**Reference Books:**


1. Wei-Meng Lee, Beginning Android 4 Application Development, Wiley Publishing, Inc.
2. Pradeep Kothari, "Android Application Development Black Book", DreamTech Press
3. James C.Sheusi, "Android Application Development for Java Programmers", Cengage Learning
4. Mark L Murphy, "Beginning Android", Wiley India Pvt Ltd
5. Sayed Y Hashimi and Satya Komatineni(2009), "Pro Android", Wiley India Pvt Ltd
6. Reto Meier, Professional Android 4 Application Development, Wiley India Pvt Ltd


**Course outcomes:**

Will be able to develop and deploy basic mobile applications.

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